

Student Name: Class:.....



YEAR 11 PRELIMINARY ASSESSMENT

TASK 1

March 2013

General Instructions

- Reading Time – 5 minutes
- Working Time – 60 Minutes
- Start each question on a new page.

Arithmetic/Surds	
Q 1	
Indices	
Q 2	
Equations	
Q 3	
Functions	
Q 4	
TOTAL	

Question 1 (10 marks)

- (a) Express $0.2\dot{3}$ as a fraction in simplest form. 2
- (b) Evaluate $\frac{\sqrt{\pi+3.4}}{\pi \times 14.9}$ correct to 3 significant figures. 2
- (c) Find integers a and b such that $\frac{4}{\sqrt{3}-2} = a + b\sqrt{3}$ 2
- (d) Rationalise the denominators and simplify $\frac{\sqrt{2}}{2\sqrt{2}+3} + \frac{2}{\sqrt{3}+1}$ 2
- (e) A merchant buys tea from a wholesaler and then sells it at a profit of 37.5%.
If the merchant sells a packet of tea for \$3.08, what price does he pay to the
wholesaler per packet of tea? 2

Question 2 Start on a new page. (14 marks)

- (a) Find in index form the value of $\frac{A^2C}{B^2}$ where $A = \left(\frac{2}{3}\right)^2$ $B = \left(\frac{4}{3}\right)^4$ $C = \left(\frac{8}{3}\right)^2$ 3
- (b) Express $\frac{3^{-1}a^{-2}}{2^{-2}a^{-3}}$ with positive indices in simplest form 2
- (c) Find the value of n and a if $(3x^n)^3 \times (3x)^{n-6} = ax^2$ 3
- (d) Simplify $\frac{16^x}{2^{3x} \times 8^{1-x}}$ 2
- (e) Simplify $\frac{a^{-2} + b^{-2}}{a^{-1}b^{-1}}$ 2
- (f) Solve $81^{2x+3} = 27^{5x-2}$ 2

Question 3. Start on a new page (14 marks)

- (a) Find the exact values of x if $2x^2 - 4x + 1 = 0$ in simplest form 2
- (b) Solve $3 - x \leq \frac{x-1}{2}$ 2
- (c) Solve $(2x-2)^2 = 18$ leaving in simplest surd form 2
- (d) Solve the following simultaneous equations
- i) $3x - y = 10$ and $x^2 + y^2 = 100$ 2
- ii) $x + y = 6$ and $xy = 8$ 2
- iii) $y = x^2 + 4x - 5$ and $y = 2x + 3$ 2
- (e) Solve $|x+2| = 2x-5$. 2

Question 4. Start on a new page (12 marks)

- (a) Sketch the graph $y = \sqrt{4-x^2}$ and state its range. 2
- (b) Sketch $y = 1 - (x+2)^2$ labelling all intercepts with the axes. 2
- (c) Sketch $y = |2x-3|$ and state the range of the function. 2
- (d) If $f(x) = x^2 + x$ express $\frac{f(x+h)-f(x)}{h}$ in simplest form. 2
- (e) Show whether the function $f(x) = \frac{x}{1-x^2}$ is odd, even or neither 2
- (f) Find the value of $g(1) - g(-2) - g(3)$ if 2

$$\begin{aligned} g(x) &= x^2 \quad \text{when } x > 2 \\ &= 2x - 1 \quad \text{when } -1 \leq x \leq 2 \\ &= 5 \quad \text{when } x < -1 \end{aligned}$$

QUESTION 1

$$a) 10x = 2.333$$

$$x = 0.2333$$

$$9x = 2.1$$

$$x = \frac{2.1}{9} = \frac{21}{90} = \frac{7}{30}$$

$$b) \frac{2.55765}{46.80973} =$$

$$= 0.0546$$

$$c) \frac{4 \times \sqrt{3} + 2}{\sqrt{3} - 2 \sqrt{3} + 2}$$

$$= \frac{4\sqrt{3} + 8}{-1}$$

$$= -4\sqrt{3} - 8$$

$$a = -8 \quad b = -4$$

$$d) \frac{\sqrt{2} \times 2\sqrt{2} - 3}{2\sqrt{2} + 3 \quad 2\sqrt{2} - 3} = \frac{4 - 3\sqrt{2}}{8 - 9}$$

$$\frac{2 \times \sqrt{3} - 1}{\sqrt{3} + 1 \quad \sqrt{3} - 1} = \frac{2\sqrt{3} - 1}{3 - 1}$$

$$\therefore -4 + 3\sqrt{2} + \frac{2\sqrt{3} - 1}{2}$$

$$= -5 + \sqrt{3} + \sqrt{2}$$

QUESTION 3

$$1) 2x^2 - 4x + 1 = 0$$

$$x = \frac{4 \pm \sqrt{16 - 8}}{4}$$

$$= \frac{4 \pm 2\sqrt{2}}{4}$$

$$= \frac{2 \pm \sqrt{2}}{2}$$

$$b) 6 - 2x \leq x - 1$$

$$7 \leq 3x$$

$$x \geq \frac{7}{3}$$

$$c) (2x - 2) = \pm \sqrt{18}$$

$$2x = 2 \pm 3\sqrt{2}$$

$$x = \frac{2 \pm 3\sqrt{2}}{2}$$

$$c) 137.5\% = 3.08$$

$$1\% = 0.0224$$

$$100\% = \$2.24$$

QUESTION 2

$$a) \left(\frac{2}{3}\right)^4 \times \left(\frac{8}{3}\right)^2$$

$$\frac{\left(\frac{4}{3}\right)^8}{\left(\frac{4}{3}\right)^8}$$

$$\frac{2^4}{3^4} \times \frac{2^6}{3^2} \times \frac{3^8}{2^{16}}$$

$$= \frac{3^2}{2^6}$$

$$b) \frac{3^{-1} a^{-2}}{2^{-2} a^{-3}} = \frac{2^2 a^3}{3 a^2} = \frac{4a}{3}$$

$$c) 3^3 \cdot x^{3n} \times 3^{n-6} \cdot x^{n-6}$$

$$\left. \begin{array}{l} 3^{n-3} = a \\ 3^{2-3} = a \\ \frac{1}{3} = a \end{array} \right\} \begin{array}{l} x^{4n-6} = x^2 \\ 4n-6 = 2 \\ 4n = 8 \\ n = 2 \end{array}$$

$$a) \frac{2^{3x} \times 2^{3-3x}}{2^{3x} \times 2^{3-3x}} = \frac{2^{3x}}{2^3}$$

$$= 2^{4x-3}$$

$$e) \left(\frac{1}{a^2} + \frac{1}{b^2}\right) \times ab$$

$$\frac{b^2 + a^2}{a^2 b^2} \times ab$$

$$= \frac{b^2 + a^2}{ab}$$

$$f) 3^{4(2x+3)} = 3^{3(5x-2)}$$

$$8x + 12 = 15x - 6$$

$$18 = 7x$$

$$\frac{18}{7} = x$$

$$d) i) x^2 + (3x - 10)^2 = 100$$

$$x^2 + 9x^2 - 60x + 100 = 100$$

$$10x^2 - 60x = 0$$

$$10x(x - 6) = 0$$

$$x = 0 \quad x = 6$$

$$y = -10 \quad y = 8$$

$$ii) x + y = 6 \quad xy = 8$$

$$x(6 - x) = 8$$

$$6x - x^2 = 8$$

$$x^2 - 6x + 8 = 0$$

$$(x - 4)(x - 2) = 0$$

$$x = 4 \quad x = 2$$

$$y = 2 \quad \text{or} \quad y = 4$$

$$iii) 2x + 3 = x^2 + 4x - 5$$

$$x^2 + 2x - 8 = 0$$

$$(x + 4)(x - 2) = 0$$

$$x = -4 \quad x = 2$$

$$y = -5 \quad y = 7$$

$$d) |x + 2| = 2x - 5$$

$$x + 2 = 2x - 5 \quad \text{or} \quad -x - 2 = 2x - 5$$

$$7 = x$$

$$-3x = -3$$

$$x = 1$$

check

$$|7 + 2| = 2 \times 7 - 5 \quad |1 + 2| = 2 \times 1 - 5$$

$$|9| = 14 - 5 \quad |3| = -3$$

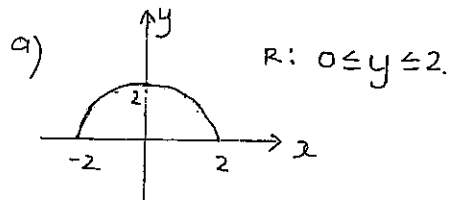
$$|9| = 9$$

False

true

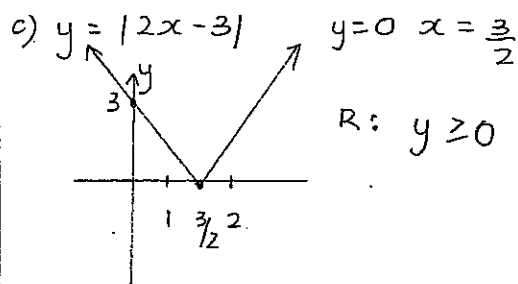
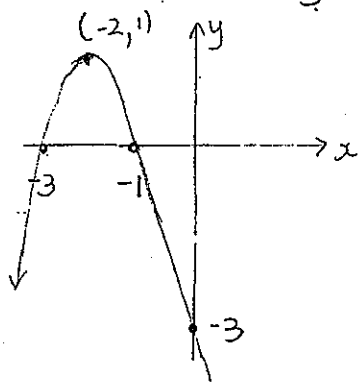
only $x = 7$ a solution

QUESTION 4



b) let $y=0$ $(x+2)^2=1$
 $x+2 = \pm 1$
 $x = -2 \pm 1$
 $= -3, -1$

let $x=0$ $y = 1 - (2)^2$
 $= -3$



d) $f(x) = x^2 + x$
 $f(x+h) = (x+h)^2 + (x+h)$
 $= x^2 + 2xh + h^2 + x + h$

$$\frac{f(x+h)-f(x)}{h} = \frac{2xh + h^2 + h}{h}$$

$$= 2x + h + 1$$

e) $f(x) = \frac{x}{1-x^2}$

$$f(-x) = \frac{-x}{1-(-x)^2}$$

$$= \frac{-x}{1-x^2}$$

$$= -f(x) \therefore \text{ODD}$$

f) $g(1) = 2x-1$
 $= 2-1$
 $= 1$

$$g(-2) = 5$$

$$g(3) = 3^2 = 9$$

$$g(1) - g(-2) - g(3)$$

$$= 1 - 5 - 9$$

$$= -13$$